Vikrant Yadav

AI Engineer (Internal AI Tooling, MCP Servers, Agentic Automation & Systematic Evals) | IIT Madras BS

South-West Delhi, New Delhi | yadavvikrant3006@gmail.com | +91 7307786930 | Portfolio | LinkedIn | GitHub

SUMMARY

Hands-on individual contributor AI Engineer and IIT Madras BS (Data Science) candidate specializing in building shared internal AI tooling, Model Context Protocol (MCP) server integrations, agentic workflow automation (Claude Code, Cursor), and versioned evaluation suites. Deeply experienced in Python, TypeScript, FastAPI, and engineering strict PHI/PII safety guardrails (spaCy NER redaction) for regulated environments. At Deep Thought Analytics, engineered reusable internal developer tooling that saved 8 to 10 team-hours daily, built asynchronous log audit tools reducing 15 to 18 hr/week tasks to 30 minutes, and deployed systems backed by **898+ automated unit and evaluation tests (97.5% MRR@10)**. Based in Delhi NCR, ready for Noida on-site operations.

EXPERIENCE

Deep Thought Analytics: Business Analyst -> AI Generalist Specialist -> Team Lead | *Nov 2025 - May 2026*

- **Internal AI Tooling and Developer Automation:** Identified recurring engineering and operational bottlenecks, building 6+ shared Python and FastAPI tools that automated company research, document extraction, and reporting (saved **~8 to 10 team-hours/day**).
- **Asynchronous Audit Tooling and Workflow Integration:** Engineered an event-driven speech and transcript analysis tool connecting Google Drive APIs to internal databases, collapsing a recurring **15 to 18 hr/week manual auditing task to ~30 minutes**
- **Deterministic Evaluation and Schema Conventions:** Implemented strict Pydantic JSON schema validation across all internal tooling to guarantee predictable input/output composition and prevent silent pipeline failures.
- **Usage Tracking and Cloud FinOps:** Tracked and monitored API token spend and model usage across internal tools on Google Cloud Vertex AI, optimizing prompt routing and embedding caches to save **INR 27,000 in cloud credits**
- **Asynchronous Technical Documentation:** Authored clear runbooks, worked examples, and SOPs for cross-functional teams, establishing reusable operating playbooks adopted independently across departments.

SKILLS

Agentic AI Tools and MCP Integrations: Claude Code, Cursor, Model Context Protocol (MCP) Servers, Project Instruction Files, Custom Tool Integrations, ReAct Agent Loops, Prompt Engineering for Reliability

Internal Tooling and Software Engineering: Python (FastAPI, Asyncio, OOP), TypeScript, Node.js, Express.js, RESTful APIs, Source Control (Git), Asynchronous Technical Documentation

Evaluation Harnesses and Observability: Tooling Evaluation Suites (Test Cases, Scoring Criteria, Pass Thresholds), RAG Triad Benchmarks, Langfuse Spend and Usage Tracking, 898+ Automated Tests (Pytest/Jest)

Security and Compliance Guardrails: Regulated Environment Awareness, PHI/PII Redaction (spaCy NER), Output Safety Guardrails, Fail-Closed Security, Version Control Discipline

Engineering Values: Self-Directed Builder, Hands-on Execution, High Data Integrity, First-Principles Architecture

PROJECTS

Multi-Agent Control Plane and MCP Server Infrastructure *(Active)* -- Python, TypeScript, LiteLLM, Docker, SQLite, Git Worktrees

- **Agentic Developer Tooling:** Architected an autonomous multi-agent developer platform where a central planner coordinates specialized tool-using worker agents running inside isolated Git worktrees and MCP server layers.
- **Versioned Evaluation and Verification Gate:** Built an automated verification gate executing test suites in sandboxed Docker containers and AST syntax audits (`tree-sitter`) before any code merge.

Behörden-Bot: Production RAG and Compliance Tooling -- Python/FastAPI, TypeScript, PostgreSQL+pgvector, LangChain | *[github.com/Vikrant*

- **PHI/PII Redaction and Fail-Closed Guardrails:** Engineered an enterprise bilingual knowledge platform featuring PII redaction (spaCy NER) and refusal guardrails to prevent data exposure in regulated contexts.
- **Systematic Evaluation Harness:** Built a Ragas-style evaluation harness integrated into CI/CD. Evaluated across **898 automated unit and integration tests**, lifting retrieval precision (MRR@10) from 75.6% to **97.5%** with sub-500ms latency.

Lead Scoring Engine: Multi-Tenant Internal Tool -- TypeScript, Express, Drizzle ORM, SQLite, Gemini | *[github.com/Vikrant038/lead-scoring-eng*

- **Structured Validation and Test Discipline:** Built a qualification engine with strict Zod schema validation and automatic model-level failovers. Maintained **317 Jest and Playwright tests** with a 90% coverage floor.

E-Commerce and Operational Analytics Tooling -- Python, Polars/Pandas, T-SQL, Streamlit, Plotly | *[github.com/Vikrant038/E-Commerce-Sales-*

- **Internal Reporting Automation:** Built an analytics platform analyzing **EUR 29.4M revenue** across 500k+ transactions with automated schema conformance testing and 72 CI-gated tests.

EDUCATION

Indian Institute of Technology Madras – BS in Data Science & Applications **|** *2025-2028* **|** CGPA: 8.67

Kendriya Vidyalaya, Pune – Senior Secondary (Class XII & X) **|** 92%

CERTIFICATIONS

Oracle Generative AI Professional | Oracle Data Science | Oracle AI Foundations | McKinsey Forward Program | Advanced Full-Stack & Python